NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

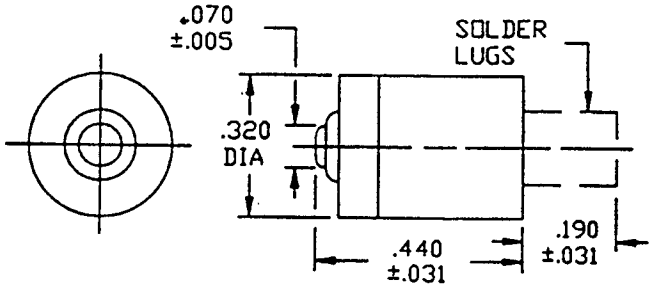
INCH-POUND
MIL-PRF-8805/68H
12 May 1995
SUPERSEDING
MIL-S-8805/68G
7 February 1992

PERFORMANCE SPECIFICATION SHEET

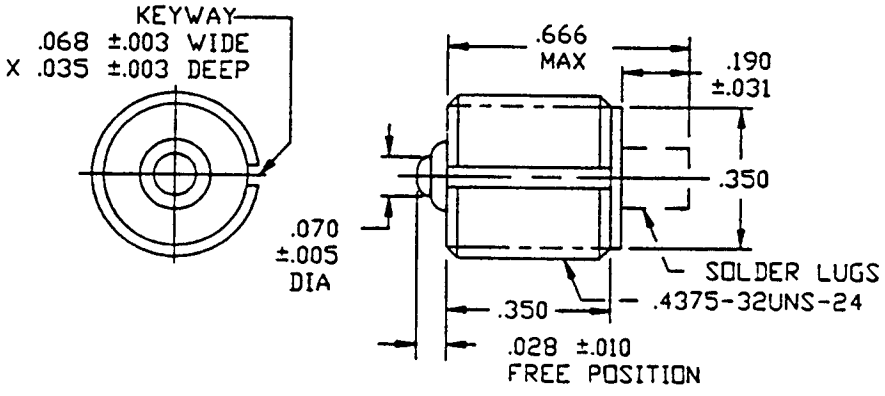
SWITCH, SENSITIVE, SPDT, 7 AMPERES AND 4 AMPERES, HERMETIC SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.



TYPE A
BASIC SWITCH



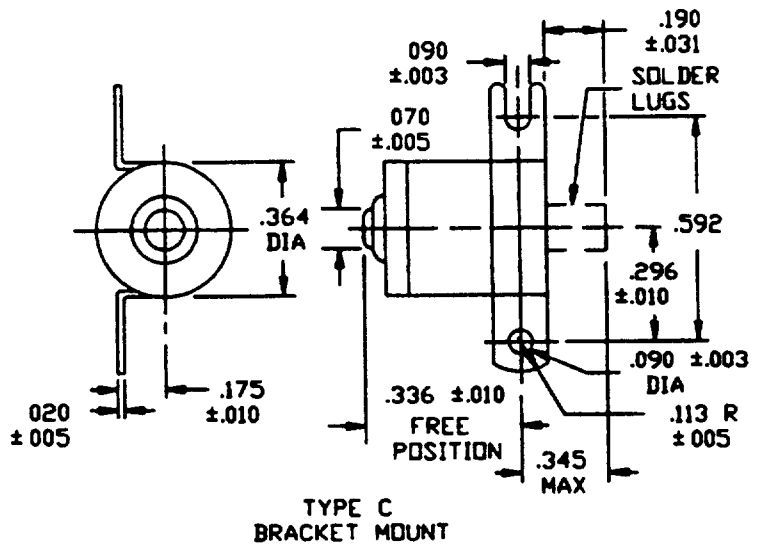
THREADED BUSHING MOUNT

FIGURE 1. Dimensions and configurations

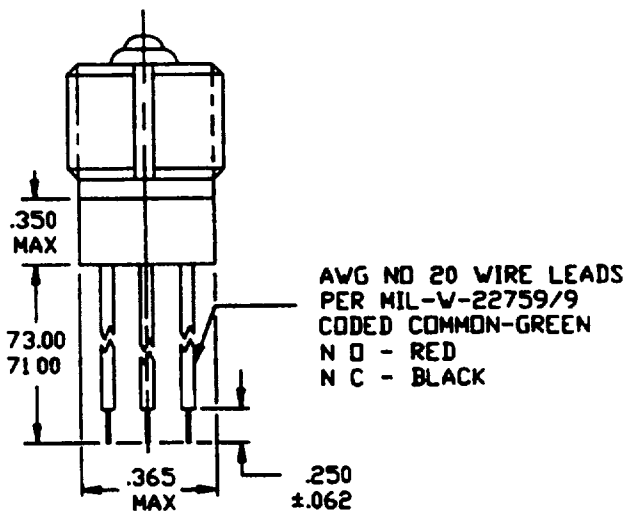
(H) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

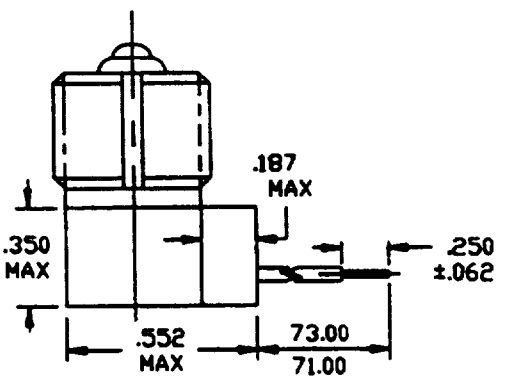
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TYPE C
BRACKET MOUNT



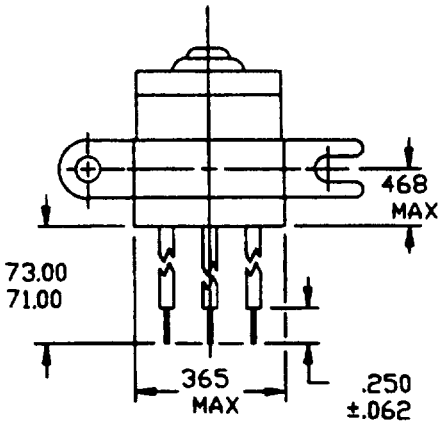
TYPE B MOUNTING
WIRE LEADS-BOTTOM



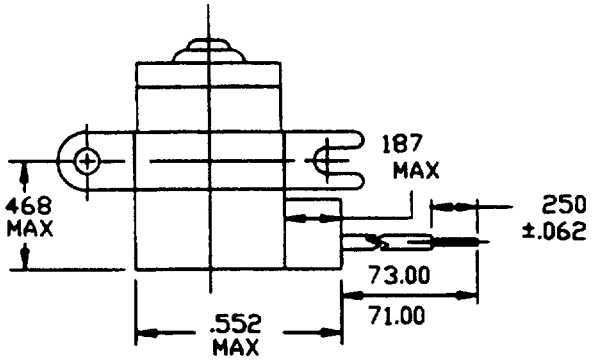
TYPE B MOUNTING
WIRE LEADS-SIDE

(H) FIGURE 1. Dimensions and configurations - continued.

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TYPE C MOUNTING
WIRE LEADS-BOTTOM



TYPE C MOUNTING
WIRE LEADS-SIDE

INCHES	MM	INCHES	MM
.003	3.83	.250	6.35
.005	0.13	.296	7.52
.010	0.25	.320	8.13
.020	0.51	.336	8.53
.028	0.71	.345	8.76
.031	0.79	.350	8.89
.035	0.89	.364	9.25
.062	1.57	.4375	11.11
.068	1.73	.440	11.18
.070	1.78	.468	11.89
.090	2.29	.592	15.04
.113	2.87	.666	16.92
.175	4.45	71.00	1803.40
.187	4.75	73.00	1854.20
.190	4.83		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±0.015 (0.38 mm).
4. Dimensions of switches with wire leads are the same as switches with solder lugs except as shown.

(H) FIGURE 1. Dimensions and configurations - continued

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 5 (hermetic). Seal test in accordance with method 112, test condition C, procedure IV of MIL-STD-202. Gross leak test condition B except water instead of silicone oil.

Temperature characteristic: 2 (-65°C to +125°C).

Shock type: M (100 G, test condition I, method 213 of MIL-STD-202).

(H) Vibration grade: 3 (10 to 3,000 Hz swept sinusoidal). In addition, the specified random vibration screening requirement is applicable to high reliability parts (PIN's M8805/68-015 through -021, -029 through -035).

(H) Materials:

PIN's M8805/68-008 through -014 and -022 through -028 (general military usage): In accordance with MIL-S-8805.

PIN's M8805/68-015 through -021 and -029 through -035 (high reliability parts): In accordance with MIL-S-8805 with additional material requirements as specified below:

Outgassing: When tested as specified in ASTM-E-595, plastic, potting compound, elastomer seals and marking ink shall meet the following outgassing requirements:

Samples to be tested shall have been processed in the same manner as that used in production of the qualification lot.

Total mass loss (TML): Shall not exceed 1.0 percent of the initial specimen mass.

Collectable volatile condensable material (CVCM): Shall not exceed 0.1 percent of the initial specimen mass.

(H) Restricted materials: Cadmium, zinc or pure tin plating shall not be used on any internal or external parts of the switch. Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent. The use of magnesium or magnesium alloyw is prohibited.

Weight:

Type A: .05 ounce maximum.

Type B: .16 ounce plus .10 ounce per lead per foot maximum.

Type C: .16 ounce plus .10 ounce per lead per foot maximum.

Operating characteristics:

Actuating force: 30 ounces maximum (at standard temperature and pressure, 21°C and 14.7 psia).

Releasing force: 4 ounces minimum (at standard temperature and pressure, 21°C and 14.7 psia).

Movement differential: .005 inch (0.13 mm) maximum.

Pretravel: .009 inch (0.23 mm) maximum.

Overtravel: .003 inch (0.08 mm) minimum.

Terminal strength: 2 pounds maximum (solder terminals only).

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Contact resistance:

Solder terminals: 40 milliohms maximum.

Wire leads: Add 11 milliohms maximum per lead per foot.

Dielectric withstanding voltage:

Sea level: 1,050 V rms terminal to case, 800 V rms terminal to terminal, 500 V rms after electrical life.

Altitude (70,000 feet).

300 V rms (solder terminals only).

1,050 V rms terminal to case (wire lead termination).

800 V rms terminal to terminal (wire lead termination).

500 V rms after electrical life.

(H) Random vibration (Applicable to PIN's M8805/68-015 through -021 and -029 through -035 only): In accordance with MIL-S-8805 and MIL-STD-202, method 214, test condition I, test condition letter B, with the following details and exceptions:

Power spectral density (PSD): +3 db/octave, 20-80 Hz; 0.04 G^2/f, 80-350 Hz flat; -3 db/octave, 350-2000 Hz. Tolerance for the PSD shall be +1.5 db, -3 db for the 20 to 1000 Hz band; +1.5 db, -4 db for the 1 kHz to 2 kHz band. The overall rms G shall be recorded for each run, and shall be within +10 percent, -15 percent of the specified curve for each acceptance.

Duration of the test shall be 1 minute minimum in each of three mutually perpendicular directions, one of which shall be in the direction of actuator movement.

Contact stability shall be continuously monitored during vibration for contact-to-contact and contact-to-case chatter. There shall be no opening or closing in excess of 10 microseconds.

Mechanical endurance: 50,000 cycles.

Electrical endurance: 25,000 cycles.

Intermediate current: 25,000 cycles.

Low level circuit: 25,000 cycles.

Electrical ratings: See table I.

(H) TABLE I. Electrical ratings.

Load	Sea level and 70,000 ft. 28 V dc	
Resistive	7	4
Inductive	1	1
Lamp	2	2

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Short circuit 20 times rated resistive load (MS3320-7 1/2 used for circuit protection).

(H) Radiographic inspection (Applicable to PIN's M8805/68-015 through -021 and -029 through -035 only):

Switches shall be radiographically inspected in accordance with method 209 of MIL-STD-202 for solder migration, loose parts and assembly anomalies. The radiographic sensitivity shall be sufficient to detect internal contamination or extraneous material equal to or greater than 0.001 inch in its major diameter. Two views shall be taken (side view and end view) in each actuator position (released and full overtravel).

(H) Run-in (Applicable to PIN's M8805/68-015 through -021 and -029 through -035 only): Each switch shall be fully activated with the following conditions:

Normally open and normally closed contacts shall be individually loaded.

(H) Contact load: .1 ampere +0, -0.1 ampere resistive.

Open circuit load voltage: 10 to 50 mV dc.

Temperature: +25°C ±5°C ambient.

Cycles and rate: 250 cycles minimum at a rate of 20 cycles per minute maximum with nominally equal "on" and "off" times.

Monitoring: The monitoring equipment shall record the number of failures or discontinue the test if a failure occurs (failure of a contact to open or close its individual circuit in proper sequence).

Qualification.
Group submission: See table II.

(H) TABLE II Qualification inspection group submission.

Inspection	Basic Switch	Additional inspections	Extent of Approval
Qualification inspection table of MIL-S-8805	M8805/68-009, -023 (all sample units) (-009 will qualify the -023 because of the load rating, but not the reverse)	M8805/68-008, 010, 011, 012, 013, 014, 022, 024, 025, 026, 027 and 028: (2 sample units each) Visual and mechanical inspection Terminal strength	All part numbers
Certification to material requirements	M8805/68-015 through -021 and -029 through -035		

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(H) Group A inspection:

M8805/68-008 through -014 and -022 through -028: In accordance with MIL-S-8805.

M8805/68-015 through -021 and -029 through -035: Each switch shall be screened in accordance with the requirments of table III in the order listed, except that the seal testing may be performed out of the specified sequence. In addition, a sample of three switches shall be taken from the screened lot and tested in accordance with table IV in the order listed.

TABLE III. Screening requirements for high reliability parts

100 percent inspection
Radiographic inspection
Run-in
Dielectric withstanding voltage
Insulation resistance
Contact resistance
Seal

Table IV. Quality conformance inspection for high reliability parts.

3 samples from each screened lot
Random vibration
Radiographic inspection
Dielectric withstanding voltage
Insulation resistance
Contact resistance
Seal

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Part or identifying number (PIN). See table V.

Substitution information See table VI.

(H) TABLE V. PIN's and configuration.

PIN	Resistive Load Rating	Type mounting	Terminals	Application
M8805/68-008	7 Amp	A	Solder	General military applications
M8805/68-009	7 Amp	B	Solder	
M8805/68-010	7 Amp	B	Wire lead - bottom	
M8805/68-011	7 Amp	B	Wire lead - side	
M8805/68-012	7 Amp	C	Solder	
M8805/68-013	7 Amp	C	Wire lead - bottom	
M8805/68-014	7 Amp	C	Wire lead - side	
M8805/68-015	7 Amp	A	Solder	High reliability parts 1/
M8805/68-016	7 Amp	B	Solder	
M8805/68-017	7 Amp	B	Wire lead - bottom	
M8805/68-018	7 Amp	B	Wire lead - side	
M8805/68-019	7 Amp	C	Solder	
M8805/68-020	7 Amp	C	Wire lead - bottom	
M8805/68-021	7 Amp	C	Wire lead - side	
M8805/68-022	4 Amp	A	Solder	General military applications
M8805/68-023	4 Amp	B	Solder	
M8805/68-024	4 Amp	B	Wire lead - bottom	
M8805/68-025	4 Amp	B	Wire lead - side	
M8805/68-026	4 Amp	C	Solder	
M8805/68-027	4 Amp	C	Wire lead - bottom	
M8805/68-028	4 Amp	C	Wire lead - side	
M8805/68-029	4 Amp	A	Solder	High reliability parts 1/
M8805/68-030	4 Amp	B	Solder	
M8805/68-031	4 Amp	B	Wire lead - bottom	
M8805/68-032	4 Amp	B	Wire lead - side	
M8805/68-033	4 Amp	C	Solder	
M8805/68-034	4 Amp	C	Wire lead - bottom	
M8805/68-035	4 Amp	C	Wire lead - side	

1/ High reliability parts are subjected to increased levels of screening and quality conformance inspection as specified in tables III and IV

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(H) TABLE VI. Canceled PIN's and substitution information. 1/

Canceled PIN's M8805/68-	7 Amp PIN's M8805/68-	4 Amp PIN's M8805/68-
001	008	022
002	009	023
003	010	024
004	011	025
005	012	026
006	013	027
007	014	028

1/ The canceled PINs are no longer acquirable. The substitute parts are interchangeable in all respects except electrical ratings. The canceled parts were rated at sea level and 70,000 feet altitude as follows:

28 V dc: 7 A resistive, 3 A inductive, 2 A lamp load.
115 V ac, 400 Hz: 5 A resistive, 3 A inductive, 2 A lamp load.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AV, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - ES

(Project 5930-1594-01)

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